The Shifting Lexicon of American Power: Computational Analysis of Political Language in State of the Union Addresses, 1950–2026

Calder Katyal
Yale University
HUMS 3210 (Culture as Data: Thinking Computationally with Text)
calder.katyal@yale.edu

Spring 2026

Abstract

This paper uses computer-based text analysis on a collection of 79 speeches given by U.S. presidents from 1950 through 2026. The collection includes all State of the Union addresses from Truman through Trump's second term. Analyzing the transformation of the SOTU language over time using TF-IDF Differential Analysis, LDA Topic Modeling, Supervised Classification (Linear SVM), and t-SNE Document Embeddings, we find that (1) the language of the SOTU has been changing dramatically over time to reflect changes in geopolitical and internal U.S. challenges; (2) there are measurable differences in the rhetoric used by Democratic and Republican presidents, with a Linear SVM classifier achieving a cross-validated accuracy of 92.4% in distinguishing party from speech; and (3) proximity in time is at least as strong an organizing principle in the embedding space as is party affiliation.

1 Introduction

The State of the Union is a unique instance in the American archive of political language, a result of Article II, Section 3 of the U.S. Constitution requiring an annual address. It is also one of the few textual instances created over such a long time span with very similar institutional requirements—a single voice (the president) addressing a relatively constant audience (Congress and now the nation through televised broadcasts) with repeated opportunities to present common messages. Due to this similarity in format, the SOTU corpus has become a potentially unique object in which to study how the political vocabulary of America has changed, and how these changes reflect relationships between rhetoric and history.

In addition to asking about changes in the vocabulary used in SOTU addresses from 1950 to 2026, this paper investigates whether there are discernible differences in the vocabulary used by Democratic versus Republican presidents. Additionally, because some of the analytical tools described in this paper were developed using large bodies of data, they may be useful in understanding how effective these methods are in detecting certain types of linguistic variations. However, due to their reliance on abstractions and generalizations, these same methods may have blind spots or limitations when identifying other forms of variation.

As such, the methodology employed here uses multiple techniques including vector semantics (TF-IDF), generative topic modeling (LDA), supervised machine learning, and dimensionality reduction for visualization. Based upon Underwood's (2019) claim that "computational methods can uncover patterns that are invisible to human readers working at the level of individual texts," each of the speeches addressed in this research is treated as a document within a collection and analyzed via distributional methods to determine if there are any significant temporal shifts or partisan divergences in terms of vocabulary usage. At the

same time, however, we take seriously the warnings provided by Drucker (2017) and D'Ignazio and Klein (2020), who point out that all data visualizations and quantification methodologies assume something about what is being counted, what can be compared, and what will be hidden from view as a result of the process of abstracting.

2 Data Curation and Preprocessing

2.1 Assembling the Corpus

Although every corpus may be interpreted differently, we believe it is necessary to describe the decision-making process behind how we created this corpus. The corpus contains 79 speeches given by 15 U.S. presidents who were in office between 1950 and 2026. We used 73 speeches from 1950 through 2020 from the `stdlib-js/datasets-sotu` dataset, which comes from a GitHub repository utilizing data from the American Presidency Project at the University of California, Santa Barbara. We also collected four speeches from President Joe Biden (2021–2024) and two from President Donald J. Trump's second term (2025–2026); these were gathered directly from the American Presidency Project (`presidency.ucsb.edu`), as complete transcripts for each speech included both the comments of the president and his audience.

Our use of two different datasets resulted in a small difference in approach. The `stdlib-js` dataset had virtually no edits made to it while the transcripts for each speech available at the Presidency Project contained stage directions (e.g., "[Laughter]", "[Applause]") in addition to statements made by the president. While we could have cleaned the datasets to eliminate the performance aspects of each document, we chose not to do so since contextual information about how certain statements were received may be lost. Since the stage direction content accounts for approximately 2% of the total number of tokens, and several document frequency thresholds would exclude most of the stage directions, retaining them likely has little-to-no effect on the results.

We chose to begin the corpus in 1950 for both pragmatic and analytical reasons. Before the mid-twentieth century, SOTU addresses were almost exclusively written messages sent to Congress—a fundamentally different genre from the televised oral address that became standard after Truman. Starting in 1950 keeps the corpus to a relatively coherent rhetorical form, even as that form evolves from radio-era formality to the applause-punctuated cadences of contemporary addresses.

Table 1: SOTU Corpus Summary by President

President	**Party**	**Years**	N
Truman	D	1950–53	4
Eisenhower	R	1953–61	9
Kennedy	D	1961–63	3
Johnson	D	1964–69	6
Nixon	R	1970–74	5
Ford	R	1975–77	3
Carter	D	1978–81	4
Reagan	R	1982–88	7
G.H.W. Bush	R	1989–92	4
Clinton	D	1993–2000	8
G.W. Bush	R	2001–08	8
Obama	D	2009–16	8
Trump (1st)	R	2017–20	4
Biden	D	2021–24	4
Trump (2nd)	R	2025–26	2
Total			**79**

In order to provide context on the nature of the data included in our corpus (Table 1), it is important to note that our sample includes 37 speeches given by Democrats and 42 speeches given by Republicans, roughly reflecting the distribution of presidential terms across this period. Each speech averaged about 6,014 words in length, but with significant variation. For example, Carter's 1981 written message had more than 33,000 words, while Nixon's 1973 written message (a formal, short written report presented at the beginning of his second term) consisted of fewer than 1,700 words. These variations in length are not just analytic, as they demonstrate the continued usage of written policy messages alongside oral oratory throughout the postwar period.

2.2 Preprocessing Pipeline

All of the speeches were first tokenized using NLTK's `word_tokenize` function. After tokenizing, all the letters in each speech were converted to lowercase. Stopwords were removed from the

content-word datasets to ensure only actual words would be analyzed, including the bare contraction fragments ("ve," "re," "ll," "nt") that NLTK's tokenizer produces from forms like *we've* and *don't*. In addition to removing stopwords, when performing TF-IDF and topic modeling we chose to incorporate both unigrams and bigrams into the analysis. We established a minimum document frequency of 3 and a maximum document frequency of 90%, determining that the dictionary size would consist of 5,000 features. The parameters selected above are typical values found in computational text analysis (Jockers, 2013), while attempting to maintain a balance between feature space and noise. Prior to removing stopwords, the average sentence length, the type-token ratio, and the moving-average type-token ratio (MATTR, with a 50-token window) of each speech were computed using NLTK's `word_tokenize` and `sent_tokenize`.

3 Methods and Results

3.1 Lexical Trends Over Time

We begin with what might be called the material history of the SOTU address: how its length and lexical texture have changed over seven decades.

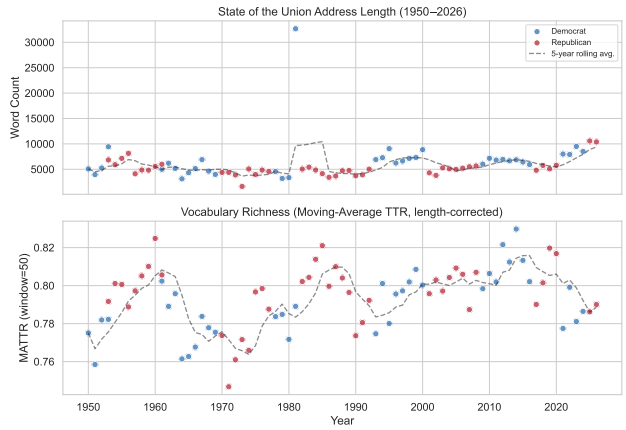

Figure 1: Speech length (word count) and length-corrected vocabulary diversity (MATTR, 50-token window), 1950–2026. Points colored by party; dashed line shows a 5-year rolling average.

There are two distinct trends apparent when examining Figure 1. The first trend is that overall length has fluctuated over time. Each decade's average length for speeches falls into a general range of approximately 4,000 to 7,000 words. Although the 1970s had the lowest average length (approximately 4,100 words), the 1980s' elevated average ($\approx$ 7,200 words) can be attributed almost entirely to the inclusion of President Carter's 32,676-word written address of 1981, delivered after he had lost re-election. A second trend apparent at the conclusion of this data collection is significant divergence. Trump's two messages during his second term (2025–2026) and Biden's later addresses (2023–2024) are among the longest in the entire corpus, ranging from roughly 8,500 to 10,600 words. Additionally, the average word count per speech for the 2020s is the greatest of any decade. Vocabulary diversity is reported using the length-corrected MATTR (moving-average TTR with a 50-token window) rather than raw TTR, because raw TTR mechanically declines with document length and the 2020s contain the longest speeches. Under MATTR, the picture is essentially flat: MATTR sits in a narrow 0.78–0.81 band across decades, and the partisan gap visible in raw TTR (Republican 0.277 vs. Democratic 0.240) nearly disappears (MATTR 0.796 vs. 0.790).

3.2 Partisan Language: Differential TF-IDF

We determined how to identify which words characterized each political party using TF-IDF (term frequency–inverse document frequency) vectorization techniques. We generated separate averaged TF-IDF feature vectors for each political party, rather than treating the individual presidential addresses as single data points. This provided us with an aggregate measure of what characterizes each political party without over-representing the longer presidential addresses of Carter and Truman. Figure 2 shows the top 20 most discriminatory words within each party's vocabulary.

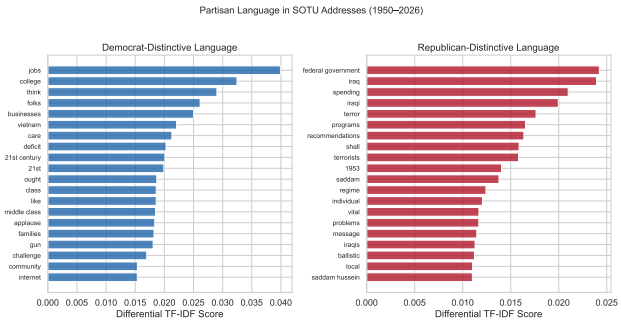

Figure 2: Top 20 terms by differential TF-IDF score for each party. Higher scores indicate stronger association with that party's speeches.

We interpret the word lists directly: we see clearly the distinct vocabularies for each political party. The list of words that characterize Republicans consists of *iraq*, *federal government*, *spending*, *terror*, and *terrorists*. It is easy to recognize that the Republican Party has long emphasized national security and limited government—ideas expressed consistently during the George W. Bush presidency and continued through subsequent Republican presidencies. On the other hand, the distinct vocabulary of Democrats is composed of the words *jobs*, *college*, *middle class*, *businesses*, *folks*, and *21st century*. Clearly, this set of words illustrates the Democratic Party's emphasis on providing greater access to economic opportunities and its use of a direct address style of communication that began with Clinton and became more prominent with Obama and Biden.

However, these differences are more than just ideological; they are also temporal. For instance, the strong association of Republicans with the terms "Iraq" and "terror" indicates that the lengthiest continuous stretch of time in which Republicans gave State of the Union addresses to Congress occurred immediately after September 11. Furthermore, while the usage of colloquialisms like "folks" by Democrats may indicate some degree of stylistic or register variation in presidential public speaking since the 1990s, it cannot represent a characteristic of their ideology. Because both party affiliation and time coexist in all analyses of this type, this represents one of several methodological challenges that remain throughout this paper.

3.3 Topic Modeling (LDA)

We used Latent Dirichlet Allocation (LDA) with $k = 8$ topics across our full body of text. Following Chang et al.'s (2009) recommendations, we labeled each topic based upon its most representative words; we did not impose additional labels after the fact.

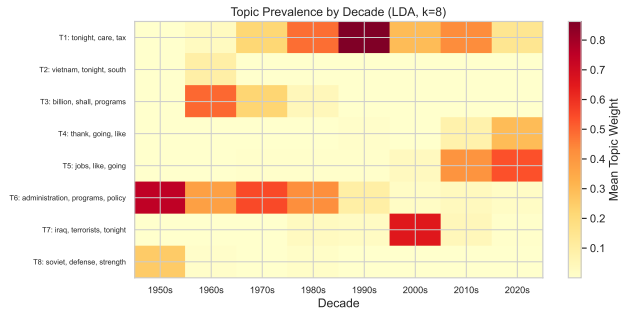

Figure 3: Mean topic prevalence by decade (LDA, $k = 8$). Rows are topics labeled by top-3 words; columns are decades.

In Figure 3, we see that topic prevalence changes dramatically over time. There was one topic that was characterized by "soviet," "defense," and "strength" words, and it concentrated in the 1950s and 1960s. This topic disappeared at an extremely fast rate once the early Cold War's vocabulary lost reference. Another topic was focused on "vietnam," "tonight," and "south." It peaked during the 1960s, but dropped off in the 1970s. A third topic had the theme "iraq," "terrorists," and "tonight," and this topic was very strongly localized to the 2000s. During the middle of the 20th century (i.e., from approximately the 1950s to the 1980s), a fourth topic ("administration," "programs," and "policy") dominated the State of the Union addresses; however, this topic collapsed in the 1990s. Finally, there was a fifth topic composed of "thank," "going," and "like," which had virtually no presence prior to the 2010s, but rose sharply in prominence beginning around that point and completely dominated all SOTUs from 2020 forward.

Therefore, the overall trend can be described as a stratigraphy of U.S. political language: from Cold War defense language, through Great Society program language, followed by post-Vietnam economic anxiety, then War on Terror language, and finally a colloquial populist tone that characterizes the majority of SOTUs since the early 2010s. Nel-

son (2010) has stated that topic models are best understood as starting points for interpretation—i.e., they provide guidance regarding where to begin close readings.

3.4 Supervised Classification

We tested whether a machine learning algorithm could determine which political party a U.S. President represented based upon the content of his annual State of the Union Address. We used four algorithms; each was given the same dataset and run with 5-fold stratified cross-validation on TF-IDF features (3,000 features, unigrams and bigrams). As shown in Table 2, the Linear SVM achieved an accuracy of 92.4%, substantially above the 53% majority-class baseline.

Table 2: Party Classification Accuracy (5-Fold CV)

Model	Mean Acc.	Std.
Logistic Regression	0.798	0.091
Linear SVM	**0.924**	0.061
Random Forest	0.823	0.060
Gradient Boosting	0.708	0.081

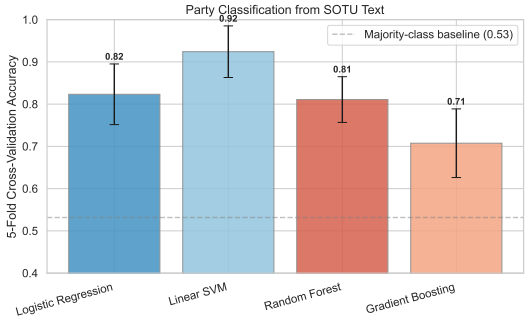

Figure 4: Classification accuracy across four models. Error bars show standard deviation across folds. Dashed line indicates the majority-class baseline (53%, given 42/79 Republican speeches).

That a linear classifier was able to achieve such high accuracy utilizing a simple bag-of-words feature set is a significant finding. It indicates that there exist enough distinguishable words in the language associated with both parties to enable a classification model to identify a speaker's party affiliation. The finding is also consistent with what So (2017) found in his research; he reported that imperfect models can still provide useful information if their behavior can be interpreted.

3.5 Document Embedding Visualization

To examine the global structure of the corpus without the mediation of a classifier, we projected all 79 speeches into two dimensions using TF-IDF features, dimensionality reduction via Truncated SVD (30 components), and t-SNE (perplexity=15).

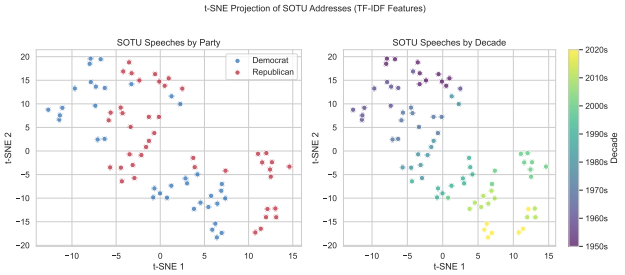

Figure 5: t-SNE projection of SOTU addresses. Left: colored by party. Right: colored by decade.

One of the most interesting visualizations in this paper appears in Figure 5. The left side of the figure shows how well the speeches clustered together based upon party affiliation. While the clusters show some separation between the two groups, there is considerable overlap within each group. On the other hand, the speeches cluster much more clearly when organized chronologically (the right side of the figure), demonstrating a clear trend across time, from speeches delivered during the 1950s located in one area of the space, to those delivered in the 2020s located in another area of the space. Moreover, many speeches fall somewhere along the line between these points over time, depending upon when they were delivered. For example, a speech delivered by Bill Clinton would likely be placed near a speech delivered by George W. Bush due to the strong commonality of the shared language used during the 1990s through 2000s, while a speech delivered by Harry Truman would have little to do with either of these speeches.

These findings support Underwood's (2019) work concerning the fact that historical periods often overpower every other type of distinction made in text-based studies, and suggest that researchers who train models to classify texts into specific cat-

egories (e.g., political party) may unintentionally be training them to recognize patterns related to time.

3.6 Keyword Trends and Sentence Complexity

We tracked the per-speech frequency of terms related to four thematic categories—security and defense, economy, social policy, and foreign policy—across the full corpus (Figure 6).

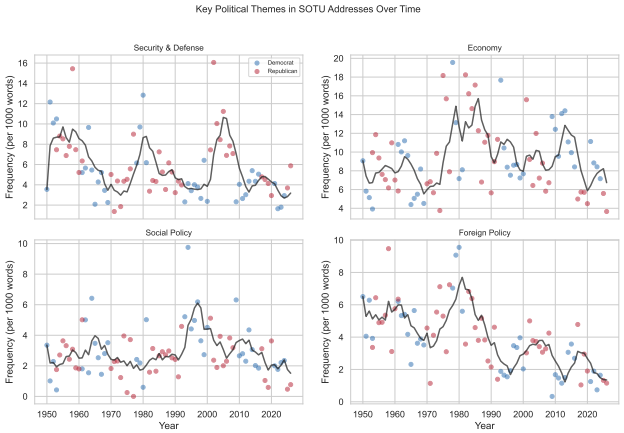

Figure 6: Frequency of key political terms (per 1,000 words) across four thematic domains, 1950–2026.

The trends are historically interpretable. Defense and security language was at its most frequent usage in the 1950s, approximately 9.0 per 1,000 words—as a result of the Korean War and the early Cold War—then decreased dramatically throughout the 1960s, had secondary high points due to the build-up of the 1980s under Reagan and the post-September 11 period, and reached its lowest corpus-wide frequency levels in the 2010s and 2020s. Economic terminology increased continuously through the 1960s and 1970s, was at its highest average value in the late Carter/Reagan era (the 1980s average was the highest for any decade), and continued to be higher than normal until the decline in economic terminology in the 2020s. Social policy terminology was at its highest value in the Clinton era (1990s), and there were additional smaller groupings of social terminology in the George W. Bush era. The Great Society addresses, contrary to what might have been expected, had only moderate values when looking at this same data. Foreign policy terminology remained relatively consistent from the 1950s through the 1980s (averaging about 5 foreign policy terms per 1,000 words); however it dropped off significantly after that point and continued to drop off over time, reaching its lowest values in the 2010s and 2020s. This is an example of a lexical representation of the shift towards domestic populist language within both parties.

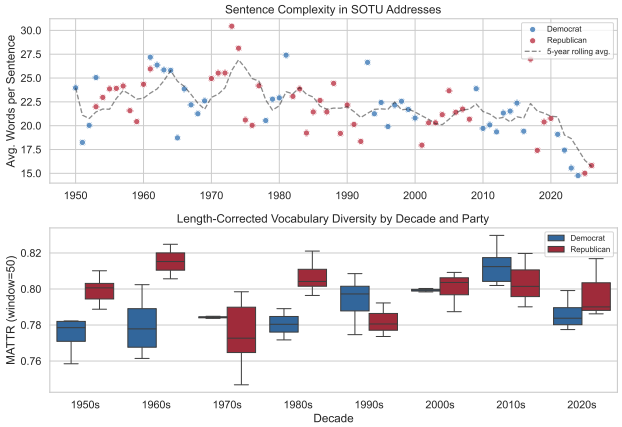

Figure 7: Sentence complexity (average words per sentence) and vocabulary diversity by decade and party.

Figure 7 presents information regarding sentence-level readability. Average sentence lengths were essentially flat from the 1950s through the 1980s (approximately averaging between 22 and 25 words per sentence) and declined after that. By the 2020s, average sentence length had fallen to approximately 17 words per sentence. As one can see in Figure 7, the decline began much later than an account based upon a "television revolution" would suggest. The lower panel uses the length-corrected MATTR for vocabulary diversity (see §3.1); under MATTR, the partisan and decadal differences are modest.

4 Discussion

Three important conclusions can be drawn from the analysis of SOTU speeches conducted here. Each conclusion has major implications for how to analyze political texts using computers.

The influence of historical time. In all analyses—regardless of whether it was topic

models, keyword tracking, t-SNE, or TF-IDF—the main structural element influencing SOTU speeches was the time period during which they were given. For example, the speeches of U.S. Presidents during the Cold War era of the 1950s are distinctly different in their linguistic style from the speeches of U.S. Presidents after 9/11 during the 2000s; and those speeches are similarly distinct from the populist styled speeches of U.S. Presidents during the 2020s. As Schöch (2013) reminds us, big data in the humanities is also messy data; there is no way to separate out the impact of time from all of the other variables you would like to explore.

There are partisan signatures—but they are not permanent. While our 92.4% classification result clearly shows that the language used by Democratic and Republican U.S. Presidents is significantly different, that difference changes based upon when those speeches occurred. The t-SNE visualization (see Figure 5) indicates that the speeches of the 1950s–1960s cluster together in chronological order rather than by party affiliation. By contrast, later speeches appear to have some degree of partisan separation. It is unclear from these results whether the increase in partisanship in recent years reflects a true increase in disagreement among politicians or if it is due to the overlap of party identity with the unique lexical styles of post-9/11 and post-2020 discourse. We flag this as an area of investigation for future research.

Computational methods have limits. Every one of the analytical techniques employed in this paper reduces SOTU speeches into word-term distributions, losing syntax, rhetorical organization, and logical argumentation—fundamentally a trade-off made by what Moretti (2013) calls "distant reading": scale for depth. According to Drucker (2017), the "distant" in distant reading is both spatial and epistemic. One possible best way to leverage these computational results will be to find where close reading should occur: i.e., where the model's confidence is highest and/or lowest.

Limitations. Our dataset is relatively small with 79 documents. Since SOTU addresses are collaborative texts that are typically written in part by speech writers for each President and party, caution is warranted when making causal attributions to either individual Presidents or political parties for patterns found. Also, since we grouped our cross-validation folds by year (and not by a particular President) some of the reported classification accuracy may have been influenced by the stylistic writing habits of the specific Presidential administration (as opposed to an actual partisan message). Future research should explore using contextually embedded models like BERT, or other Large Language Model based methods for evaluating topics (Yang et al., 2024).

5 Conclusion

Returning to the three questions we asked at the beginning: the vocabulary used for State of the Union (SOTU) speeches has changed considerably from 1950 through 2026. These changes are consistent with what was happening during the time frames of each era. Although there are detectable differences in the way Democrats and Republicans talk in SOTU speeches, this difference is not constant or predictable over history. Computational techniques have been able to identify much of the overall structure in how SOTU speech has evolved over the years. Therefore, the evidence supports the main argument: it appears that the State of the Union address is influenced more by when a president was elected than by who he represents.

References

Chang, J., Boyd-Graber, J., Gerrish, S., Wang, C., and Blei, D. M. (2009). Reading tea leaves: How humans interpret topic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 288–296.

D'Ignazio, C. and Klein, L. F. (2020). Feminist data visualization. In *Data Feminism*. MIT Press, Cambridge, MA.

Drucker, J. (2017). Why distant reading isn't. In *PMLA*, volume 132, pages 628–635.

Jockers, M. L. (2013). *Macroanalysis: Digital Methods and Literary History*. University of Illinois Press, Urbana.

Moretti, F. (2013). *Distant Reading*. Verso, London.

Nelson, R. K. (2010). Mining the dispatch. *Digital Scholarship Lab, University of Richmond*.

Schöch, C. (2013). Big? smart? clean? messy? data in the humanities. In *Journal of Digital Humanities*, volume 2.

So, R. J. (2017). All models are wrong. *PMLA*, 132(3):668–673.

Underwood, T. (2019). *Distant Horizons: Digital Evidence and Literary Change*. University of Chicago Press, Chicago.

Yang, X., Zhao, H., Phung, D., Buntine, W., and Du, L. (2024). LLM reading tea leaves: Automatically evaluating topic models with large language models. *arXiv preprint arXiv:2406.09008*.